

Incident Escalation & On-Call Rotation Engine

Aryan N – PES1UG24CSS656

Written Justification – Architectural Pattern Selection

Architecture Selection: “We chose Microservices Architecture for the Incident Escalation & On-Call Rotation Engine.”

Architectural Choice

The system is decomposed into six independently deployable services: Alert Ingestion Service, On-Call Rotation Service, Escalation Engine Service, Notification Dispatch Service, Incident Tracking Service, and Incident & Post-Mortem Store. Each exposes a well-defined interface (IAlertIntake, IRotationLookup, INotify, IIncidentUpdate, IIncidentQuery, IIncidentPersistence) and the services collaborate over API calls, with additional «use» dependencies where one service invokes another as part of a workflow (e.g. Alert Ingestion handing off to the Escalation Engine after assigning the primary engineer).

Two Reasons for the Choice

1. Independent scaling and fault isolation for the notification path. FR-003 and NFR-001 require alerts to be dispatched via SMS, phone, or email within 3 seconds even under peak load. Because Notification Dispatch is its own service, it can be scaled out independently of Alert Ingestion, On-Call Rotation, or Incident Tracking, and if an SMS/phone gateway is slow or fails, that failure stays isolated instead of blocking alert intake or rotation lookups — directly supporting NFR-002's requirement that no incident or escalation event be lost.
2. Independent deployability for frequently-changing escalation logic. FR-002's 5-minute acknowledgement/escalation rule is the part of the system most likely to change (timeout values, retry counts, new escalation policies). Keeping the Escalation Engine as a separate service means it can be modified and redeployed without touching the Rotation Service, Notification Dispatch, or the Incident/Post-Mortem data store, and each service's data store or technology can evolve independently (e.g. a fast in-memory store for active timers versus a durable relational store for FR-005's post-mortem records).

Security Advantage

Each service can be secured with least-privilege access to only the credentials it needs. The Alert Ingestion Service is the externally-facing endpoint that receives webhook alerts from the Monitoring System, but it never holds notification-provider credentials or database credentials directly — it only calls the On-Call Rotation Service through IRotationLookup. SMS/phone/email provider credentials are confined to the Notification Dispatch Service, and access to stored incident and post-mortem records is confined to the Incident Tracking Service through IIncidentPersistence. If the internet-facing ingestion endpoint were ever compromised, the attacker would not automatically gain access to notification channels or the incident data store.

Performance Benefit

Because Alert Ingestion and Notification Dispatch are separate, independently scalable services, additional instances of just these two services can be started during a spike in monitoring alerts to

keep dispatch initiation under the 3-second target set by NFR-001, without needlessly over-provisioning the On-Call Rotation Service or Incident Tracking Service, which are read-mostly and far lighter-weight by comparison.

Trade-off Accepted

Splitting the system into services introduces network latency between calls and operational complexity, such as keeping the Incident Tracking Service and the Incident & Post-Mortem Store consistent during high escalation volume. This is an acceptable trade-off because fault isolation and independent scaling are critical for a 24/7 on-call system, where one failing channel (e.g. an SMS provider outage) must never be allowed to stop the whole alerting and escalation pipeline.